

SUSTech

Midterm for Calculus II in Spring Semester, 2019

注意：本试卷里的中文为直译（即完全按英文字面意思直接翻译），所有数学词汇的定义请参照教材（Thomas' Calculus, 13th Edition）中的定义。如果其中有些数学词汇的定义不同于中文书籍（比方说同济大学的高等数学教材）里的定义，以教材（Thomas' Calculus, 13th Edition）中的定义为准。

1. (15 pts) Determine whether the following statements are **true** or **false**? No justification is necessary.

(1) If $a_n < 0$ for $n > 1000$, and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ may diverge.

(2) Suppose that the power series $\sum_{n=1}^{\infty} a_n(x-1)^n$ converges at $x=0$, and diverges at $x=2$, then the interval of convergence of this series is $[0, 2)$.

(3) If $f(x, y)$ has partial derivatives $f_x(x, y)$ and $f_y(x, y)$ at (x_0, y_0) , then

$$\lim_{x \rightarrow x_0} f(x, y_0) = \lim_{y \rightarrow y_0} f(x_0, y) = f(x_0, y_0).$$

(4) If a vector function $\mathbf{r}(t)$ is always perpendicular to its derivative $\frac{d\mathbf{r}}{dt}$, then $|\mathbf{r}(t)|$ must be constant.

(5) The curvature of a unit circle is greater than the curvature of the parabola $y = x^2$ at the origin.

一、（15分）判断题：

(1) 若当 $n > 1000$ 时 $a_n < 0$ ，并且已知 $\sum_{n=1}^{\infty} a_n$ 收敛，则 $\sum_{n=1}^{\infty} a_n^2$ 有可能发散。

(2) 若幂级数 $\sum_{n=1}^{\infty} a_n(x-1)^n$ 在 $x=0$ 处收敛，而在 $x=2$ 处发散，那么该级数的收敛区间是 $[0, 2)$ 。

(3) 若函数 $f(x, y)$ 在 (x_0, y_0) 处存在偏导数 $f_x(x, y)$ 和 $f_y(x, y)$ ，那么

$$\lim_{x \rightarrow x_0} f(x, y_0) = \lim_{y \rightarrow y_0} f(x_0, y) = f(x_0, y_0).$$

(4) 若一个向量函数 $\mathbf{r}(t)$ 总是垂直于其导数 $\frac{d\mathbf{r}}{dt}$ ，则 $|\mathbf{r}(t)|$ 一定是常数。

(5) 在原点处，单位圆的曲率大于抛物线 $y = x^2$ 的曲率。

2. (9pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

- (1) Let $\mathbf{a}$ and $\mathbf{b}$ be two nonzero orthogonal vectors, which of the following must be true?
 (A) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$; (B) $|\mathbf{a} - \mathbf{b}| = |\mathbf{a}| - |\mathbf{b}|$;
 (C) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a} - \mathbf{b}|$; (D) $\mathbf{a} + \mathbf{b} = \mathbf{a} - \mathbf{b}$.
- (2) The equations of two lines are $l_1 : x = t, y = 2t, z = -t$, and $l_2 : x = 1 - 2t, y = t, z = -1 + t$. Then l_1 and l_2 are
 (A) parallel; (B) orthogonal; (C) intersect with each other; (D) skew.
- (3) Suppose $0 \leq a_n < \frac{1}{n}$, ($n = 1, 2, \dots$), then which of the following series must converge?
 (A) $\sum_{n=1}^{\infty} a_n$; (B) $\sum_{n=1}^{\infty} (-1)^n a_n$; (C) $\sum_{n=1}^{\infty} \sqrt{a_n}$; (D) $\sum_{n=1}^{\infty} (-1)^n a_n^2$.

二、(9分) 单项选择题:

- (1) 若向量 $\mathbf{a}$ 和 $\mathbf{b}$ 都是非零正交向量, 下面哪一个结论是正确的?
 (A) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$; (B) $|\mathbf{a} - \mathbf{b}| = |\mathbf{a}| - |\mathbf{b}|$;
 (C) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a} - \mathbf{b}|$; (D) $\mathbf{a} + \mathbf{b} = \mathbf{a} - \mathbf{b}$.
- (2) 设两条直线的参数方程为 $l_1 : x = t, y = 2t, z = -t$, 以及 $l_2 : x = 1 - 2t, y = t, z = -1 + t$. 那么 l_1 与 l_2 的关系为
 (A) 平行; (B) 正交; (C) 相交; (D) 异面.
- (3) 若 $0 \leq a_n < \frac{1}{n}$, ($n = 1, 2, \dots$), 那么下面哪一个级数一定收敛?
 (A) $\sum_{n=1}^{\infty} a_n$; (B) $\sum_{n=1}^{\infty} (-1)^n a_n$; (C) $\sum_{n=1}^{\infty} \sqrt{a_n}$; (D) $\sum_{n=1}^{\infty} (-1)^n a_n^2$.

3. (9 pts) Does the following series absolutely converge, conditionally converge, or diverge? Give reasons for your answer.

- (1) $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}$.
- (2) $\sum_{n=1}^{\infty} (-1)^n e^{-\frac{1}{n}}$.
- (3) $\sum_{n=2019}^{\infty} (-1)^n \frac{1}{\sqrt{n^2 - 2019n + 1}}$.

三、(9分) 下列级数是否绝对收敛、条件收敛或者发散? 给出你的理由.

- (1) $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}$.
- (2) $\sum_{n=1}^{\infty} (-1)^n e^{-\frac{1}{n}}$.

$$(3) \sum_{n=2019}^{\infty} (-1)^n \frac{1}{\sqrt{n^2 - 2019n + 1}}.$$

4. (10 pts)

(1) Find the radius and interval of convergence of the series

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n x^n}{\sqrt{n^2 + n + 1}}.$$

(2) For what values of x does the series converge absolutely, or conditionally?

四、 (10分)

(1) 求级数

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n x^n}{\sqrt{n^2 + n + 1}}$$

的收敛半径和收敛区间.

(2) x 取哪些值时级数绝对收敛, 取哪些值时级数条件收敛?

5. (10 pts) Let $x = \cos^3 t$, $y = \sin^3 t$, where $0 \leq t \leq \frac{\pi}{2}$, be a parametrization of a curve.

(1) Find the length of the curve.

(2) Find the area of the surface generated by revolving the curve about the x -axis.

五、 (10分) 设 $x = \cos^3 t$, $y = \sin^3 t$, $0 \leq t \leq \frac{\pi}{2}$, 为一参数化曲线.

(1) 求曲线的弧长.

(2) 求曲线绕 x 轴旋转所得到的曲面的面积.

6. (10 pts) Find the equation of the plane through the points $(2, -1, -1)$ and $(1, 0, -1)$ perpendicular to the plane $2x + 3y - 5z + 6 = 0$.

六、 (10分) 如果一个平面通过两个点 $(2, -1, -1)$ 和 $(1, 0, -1)$, 并且与平面 $2x + 3y - 5z + 6 = 0$ 垂直, 请写出平面方程.

7. (10 pts) A particle is located at the point $(1, 0, -2)$. Its initial speed is $|\mathbf{v}(0)| = 3$ at time $t = 0$, and the direction of its initial velocity is toward the point $(2, -1, 3)$. The particle moves with constant acceleration $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. Find its position vector $\mathbf{r}(t)$ at time t .

七、 (10分) 一个物体在 $t = 0$ 时初始坐标为 $(1, 0, -2)$. 它在 $t = 0$ 时的初始速率为 $|\mathbf{v}(0)| = 3$, 而且其初始速度的方向是指向点 $(2, -1, 3)$. 这个物体运动的加速度为常数向量, $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. 请写出其位移向量 $\mathbf{r}(t)$ 关于时间 t 的参数表示.

8. (10 pts) Is the following function, $f(x, y)$ continuous at $(0, 0)$? Give reasons for your answer.

$$f(x, y) = \begin{cases} \frac{\sin(x^3+y^3)}{x^2+y^2}, & (x, y) \neq (0, 0); \\ 0, & (x, y) = (0, 0). \end{cases}$$

八、（10分）请问如下函数 $f(x, y)$ 在原点处是否连续，给出你的理由.

$$f(x, y) = \begin{cases} \frac{\sin(x^3+y^3)}{x^2+y^2}, & (x, y) \neq (0, 0); \\ 0, & (x, y) = (0, 0). \end{cases}$$

9. (9 pts) Let $f(u)$ be differentiable, $z = f(e^x y)$ ($y \neq 0$), and $\frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 1$. If $f(1) = 0$, find $f(u)$.

九、（9分）设函数 $f(u)$ 可导，定义 $z = f(e^x y)$ ($y \neq 0$)，而且有 $\frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 1$. 已知 $f(1) = 0$ ，请写出 $f(u)$ 的表达式.

10. (8 pts) Find the Taylor series for $f(x) = \ln(x + \sqrt{x^2 + 1})$ at $x = 0$.

十、（8分）请写出函数 $f(x) = \ln(x + \sqrt{x^2 + 1})$ 在 $x = 0$ 处的Taylor级数.

1. (15 pts) Determine whether the following statements are **true** or **false**? No justification is necessary.

F (1) If $a_n < 0$ for $n > 1000$, and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ may diverge.

$$\sum_{n=1}^{\infty} |a_n| \text{ converges} \quad \lim_{n \rightarrow \infty} a_n = 0$$

$$n > N \quad |a_n| > a_n^2 \Rightarrow \text{Comparison Test,} \\ \sum_{n=1}^{\infty} a_n^2 \text{ converges}$$

T (2) Suppose that the power series $\sum_{n=1}^{\infty} a_n(x-1)^n$ converges at $x=0$, and diverges at $x=2$, then the interval of convergence of this series is $[0, 2)$.

Convergence Theorem for Power Series.

T (3) If $f(x, y)$ has partial derivatives $f_x(x, y)$ and $f_y(x, y)$ at (x_0, y_0) , then

$$\lim_{x \rightarrow x_0} f(x, y_0) = \lim_{y \rightarrow y_0} f(x_0, y) = f(x_0, y_0).$$

$$\begin{aligned} \forall y \quad \lim_{x \rightarrow x_0} \frac{f(x, y) - f(x_0, y)}{\Delta x} &= f_x(x, y) \\ \forall x \quad \lim_{y \rightarrow y_0} \frac{f(x, y) - f(x, y_0)}{\Delta y} &= f_y(x, y) \end{aligned} \Rightarrow \begin{aligned} \lim_{x \rightarrow x_0} f(x, y) &= f(x_0, y) \\ \lim_{y \rightarrow y_0} f(x, y) &= f(x, y_0) \end{aligned}$$

T (4) If a vector function $\mathbf{r}(t)$ is always perpendicular to its derivative $\frac{d\mathbf{r}}{dt}$, then $|\mathbf{r}(t)|$ must be constant.

$$2\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0$$

$$r^2 = c$$

$$|r|^2 = k$$

F (5) The curvature of a unit circle is greater than the curvature of the parabola $y = x^2$ at the origin.

$$y = x^2 \quad k(x) = \frac{|y''(x)|}{(1 + f'(x)^2)^{3/2}} \quad k(0) = \frac{|2|}{(1+0)^{3/2}} = 2$$

$$k_{\text{cir}} = 1$$

2. (9pts) Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) Let $\mathbf{a}$ and $\mathbf{b}$ be two nonzero orthogonal vectors, which of the following must be true?

- (A) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$; (B) $|\mathbf{a} - \mathbf{b}| = |\mathbf{a}| - |\mathbf{b}|$;
(C) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a} - \mathbf{b}|$; (D) $\mathbf{a} + \mathbf{b} = \mathbf{a} - \mathbf{b}$.

$|\mathbf{a} + \mathbf{b}|$



(2) The equations of two lines are $l_1: x = t, y = 2t, z = -t$, and $l_2: x = 1 - 2t, y = t, z = -1 + t$. Then l_1 and l_2 are

- (A) parallel; (B) orthogonal; (C) intersect with each other; (D) skew.

$$l_1: \vec{d}_1 = (1, 2, -1)$$

$$l_2: \vec{d}_2 = (-2, 1, 1)$$

$$\begin{cases} t = 1 - 2t \\ 2t = t \\ -t = -1 + t \end{cases} \Rightarrow t \text{ no solution.}$$

(3) Suppose $0 \leq a_n < \frac{1}{n}$, ($n = 1, 2, \dots$), then which of the following series must converge?

- (A) $\sum_{n=1}^{\infty} a_n$; (B) $\sum_{n=1}^{\infty} (-1)^n a_n$; (C) $\sum_{n=1}^{\infty} \sqrt{a_n}$; (D) $\sum_{n=1}^{\infty} (-1)^n a_n^2$.

$$(A) \quad a_n = \frac{1}{n \ln n}$$

$$(B) \quad a_n = \begin{cases} \frac{1}{2^n} & n = 2k \\ \frac{1}{n^2} & n = 2k+1 \end{cases}$$

$$(C) \quad a_n = \frac{1}{n^2}$$

$$(D) \quad 0 < a_n^2 < \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} (-1)^n a_n^2 < \sum_{n=1}^{\infty} a_n^2 < \infty$$

3. (9 pts) Does the following series absolutely converge, conditionally converge, or diverge? Give reasons for your answer.

$$(1) \sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}.$$

$$(2) \sum_{n=1}^{\infty} (-1)^n e^{-\frac{1}{n}}.$$

$$(3) \sum_{n=2019}^{\infty} (-1)^n \frac{1}{\sqrt{n^2 - 2019n + 1}}.$$

$$(1) \lim_{n \rightarrow \infty} \sqrt[n]{|(-1)^n (1 - \frac{1}{n})^{n^2}|} = \lim_{n \rightarrow \infty} (1 - \frac{1}{n})^n = e^{-1} < 1$$

$$\sum_{n=1}^{\infty} (-1)^n (1 - \frac{1}{n})^{n^2} \text{ converges absolutely.}$$

(2) Consider $(-1)^{2n} e^{-\frac{1}{2n}} = e^{-\frac{1}{2n}}$

$\lim_{n \rightarrow \infty} e^{-\frac{1}{2n}} = 1 \Rightarrow (-1)^n e^{-\frac{1}{n}}$ does not converges to 0.
 $\sum_{n=1}^{\infty} (-1)^n e^{-\frac{1}{n}}$ diverges

(3) $\sum_{n=2019}^{\infty} (-1)^n \frac{1}{\sqrt{n^2 - 2019n + 1}}$
 $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2 - 2019n + 1}} = 0$

$f(x) = x^2 - 2019x + 1$

$f'(x) = 2x - 2019$

$f'(x) > 0 \Rightarrow x > \frac{2019}{2}$

$n \geq 2019 \quad f(n) \geq f(2019) = 1 > 0$

$f(n+1) > f(n)$
 $\sqrt{f(n+1)} > \sqrt{f(n)} \Rightarrow \frac{1}{\sqrt{n^2 - 2019n + 1}}$ is non increasing
 $\frac{1}{\sqrt{f(n+1)}} < \frac{1}{\sqrt{f(n)}}$

By Alternative Series Test,
 $\sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n^2 - 2019n + 1}}$ converges

$\lim_{n \rightarrow \infty} \frac{\frac{1}{\sqrt{n^2 - 2019n + 1}}}{\frac{1}{n}} = 1 \quad \sum_{n=2019}^{\infty} \frac{1}{n}$ diverges

$\Rightarrow$ By Limit Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2 - 2019n + 1}}$ diverges
 $\sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n^2 - 2019n + 1}}$ converges conditionally.

4. (10 pts)

(1) Find the radius and interval of convergence of the series

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n x^n}{\sqrt{n^2+n+1}}$$

$$a_n = \frac{(-1)^{n+1} 2^n}{\sqrt{n^2+n+1}}$$

(2) For what values of x does the series converge absolutely, or conditionally?

Sol. (1) $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n x^n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{(-1)^{n+1} 2^n}{\sqrt{n^2+n+1}} \right| |x|^n} = \lim_{n \rightarrow \infty} \frac{2}{\sqrt[n]{n^2+n+1}} |x|$

$$1 \leq \sqrt[n]{n^2+n+1} \leq \sqrt[n]{4n^2} \quad \lim_{n \rightarrow \infty} \sqrt[n]{4n^2} = \lim_{n \rightarrow \infty} \sqrt[n]{4n} \cdot \sqrt[n]{2n} = 1$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n^2+n+1} = 1$$

So $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 2|x| \quad R = \frac{1}{2}$

$$2|x| < 1 \Rightarrow x \in (-\frac{1}{2}, \frac{1}{2})$$

$$\sum_{n=0}^{\infty} a_n x^n \text{ converges absolutely in } (-\frac{1}{2}, \frac{1}{2})$$

$$x = \frac{1}{2} \quad \sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n 2^{-n}}{\sqrt{n^2+n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^2+n+1}}$$

$$x = -\frac{1}{2} \quad \sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n (-2)^{-n}}{\sqrt{n^2+n+1}} = \sum_{n=0}^{\infty} \frac{-1}{\sqrt{n^2+n+1}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2+n+1}} = 0$$

$\Rightarrow$ By Alternative Series Test

for $n \geq 0$ $0 \leq \frac{1}{\sqrt{(n+1)^2+(n+1)+1}} \leq \frac{1}{\sqrt{n^2+n+1}}$

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^2+n+1}} \text{ converges}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n}} = 1$$

$\Rightarrow$ By Limit Comparison Test,

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges}$$

$$\sum_{n=0}^{\infty} \frac{1}{\sqrt{n^2+n+1}} \text{ diverges}$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^2+n+1}} \text{ converges conditionally at } x = \frac{1}{2}$$

$$\sum_{n=0}^{\infty} \frac{-1}{\sqrt{n^2+n+1}} \text{ diverges}$$

The radius of convergence is $\frac{1}{2}$.

The interval of convergence is $(-\frac{1}{2}, \frac{1}{2}]$.

(b) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n x^n}{\sqrt{n^2+n+1}}$ converges absolutely in $(-\frac{1}{2}, \frac{1}{2})$.

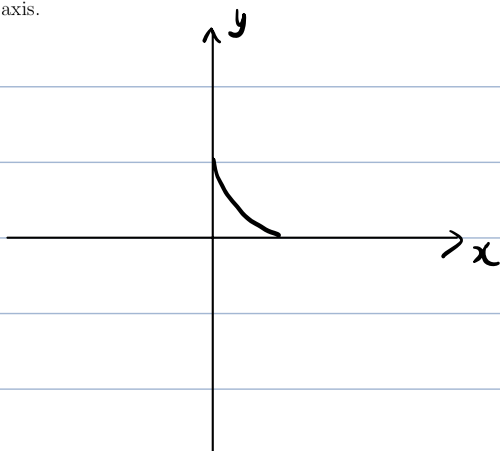
converges conditionally at $x = \frac{1}{2}$

5. (10 pts) Let $x = \cos^3 t$, $y = \sin^3 t$, where $0 \leq t \leq \frac{\pi}{2}$, be a parametrization of a curve.

(1) Find the length of the curve.

(2) Find the area of the surface generated by revolving the curve about the x -axis.

Sol.



$$(1) \quad \frac{dx}{dt} = -3\cos^2 t \sin t \quad \frac{dy}{dt} = 3\sin^2 t \cos t$$

$$\begin{aligned} L &= \int_0^{\frac{\pi}{2}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^{\frac{\pi}{2}} \sqrt{9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t} dt \\ &= \int_0^{\frac{\pi}{2}} \sqrt{9\sin^2 t \cos^2 t} dt \\ &= \int_0^{\frac{\pi}{2}} 3\sin t \cos t dt \\ &= \int_0^{\frac{\pi}{2}} \frac{3}{2} \sin 2t dt \\ &= -\frac{3}{4} \cos 2t \Big|_0^{\frac{\pi}{2}} = \frac{3}{2} \end{aligned}$$

$$\begin{aligned} (2) \quad S &= \int_0^{\frac{\pi}{2}} 2\pi y(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= \int_0^{\frac{\pi}{2}} 2\pi \sin^3 t \cdot 3\sin t \cos t dt \\ &= \int_0^{\frac{\pi}{2}} 6\pi \sin^4 t \cos t dt \\ &= \frac{6}{5} \pi \sin^5 t \Big|_0^{\frac{\pi}{2}} \\ &= \frac{6\pi}{5} \end{aligned}$$

6. (10 pts) Find the equation of the plane through the points $(2, -1, -1)$ and $(1, 0, -1)$ perpendicular to the plane $2x + 3y - 5z + 6 = 0$.

Sol. Suppose the equation of the plane S

$$ax + by + cz = k.$$

S pass through $(2, -1, -1)$ and $(1, 0, -1)$

$$\Rightarrow \begin{aligned} 2a - b - c &= k \\ a - c &= k \end{aligned}$$

The normal of S : $\vec{n}_1 = (a, b, c)$

The normal of $2x + 3y - 5z + 6 = 0$:

$$\vec{n}_2 = (2, 3, -5)$$

$$\vec{n}_1 \cdot \vec{n}_2 = 0$$

$$\Rightarrow 2a + 3b - 5c = 0$$

$$\text{We have } \begin{cases} 2a - b - c = k \\ a - c = k \\ 2a + 3b - 5c = 0 \end{cases} \Rightarrow \begin{cases} k = 0 \\ c = a \\ b = a \end{cases}$$

$$ax + ay + az = 0$$

$a = 0$ $a = b = c = 0$ We cannot get the plane

$$a \neq 0 \quad x + y + z = 0.$$

7. (10 pts) A particle is located at the point $(1, 0, -2)$. Its initial speed is $|\mathbf{v}(0)| = 3$ at time $t = 0$, and the direction of its initial velocity is toward the point $(2, -1, 3)$. The particle moves with constant acceleration $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. Find its position vector $\mathbf{r}(t)$ at time t .

$$t = 0 \quad \mathbf{r}(0) = \mathbf{i} - 2\mathbf{k} \quad \mathbf{a}(t) = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$$

Direction of $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k} - (\mathbf{i} - 2\mathbf{k}) = \mathbf{i} - \mathbf{j} + 5\mathbf{k}$

$$\mathbf{v}(0) = |\mathbf{v}(0)| \frac{\mathbf{i} - \mathbf{j} + 5\mathbf{k}}{\sqrt{1^2 + (-1)^2 + 5^2}} = 3 \frac{\mathbf{i} - \mathbf{j} + 5\mathbf{k}}{\sqrt{27}} = \frac{\sqrt{3}}{3}\mathbf{i} - \frac{\sqrt{3}}{3}\mathbf{j} + \frac{5\sqrt{3}}{3}\mathbf{k}$$

$$\mathbf{v}(t) = \mathbf{v}(0) + \mathbf{a}(t)t = \frac{\sqrt{3}}{3}\mathbf{i} - \frac{\sqrt{3}}{3}\mathbf{j} + \frac{5\sqrt{3}}{3}\mathbf{k} + t\mathbf{i} + 2t\mathbf{j} + t\mathbf{k}$$

Take the integral at both sides:

$$\mathbf{r}(t) = \frac{\sqrt{3}}{3}t\mathbf{i} - \frac{\sqrt{3}}{3}t\mathbf{j} + \frac{5\sqrt{3}}{3}t\mathbf{k} + \frac{1}{2}t^2\mathbf{i} + t^2\mathbf{j} + \frac{1}{2}t^2\mathbf{k} + \mathbf{C}$$

$$t = 0 \quad \mathbf{r}(0) = \mathbf{C} \Rightarrow \mathbf{C} = \mathbf{i} - 2\mathbf{k}$$

$$r(t) = \left(\frac{1}{2}t^2 + \frac{\sqrt{3}}{7}t + 1\right)i + \left(t^2 - \frac{\sqrt{3}}{7}t\right)j + \left(\frac{1}{2}t^2 + \frac{5\sqrt{3}}{7}t - 2\right)k.$$

8. (10 pts) Is the following function, $f(x, y)$ continuous at $(0, 0)$? Give reasons for your answer.

$$f(x, y) = \begin{cases} \frac{\sin(x^3 + y^3)}{x^2 + y^2}, & (x, y) \neq (0, 0); \\ 0, & (x, y) = (0, 0). \end{cases}$$

Pf. Yes

$$\left| \frac{\sin(x^3 + y^3)}{x^2 + y^2} \right| \leq \frac{|x^3 + y^3|}{x^2 + y^2} \leq \frac{x^2 \sqrt{x^2 + y^2} + y^2 \sqrt{x^2 + y^2}}{x^2 + y^2} = \sqrt{x^2 + y^2}$$

$$\lim_{(x, y) \rightarrow (0, 0)} \left| \frac{\sin(x^3 + y^3)}{x^2 + y^2} \right| \leq \lim_{(x, y) \rightarrow (0, 0)} \sqrt{x^2 + y^2} = 0$$

$$\text{So } \lim_{(x, y) \rightarrow (0, 0)} \frac{\sin(x^3 + y^3)}{x^2 + y^2} = 0$$

$f(x, y)$ is continuous at $(0, 0)$.

9. (9 pts) Let $f(u)$ be differentiable, $z = f(e^x y)$ ($y \neq 0$), and $\frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 1$. If $f(1) = 0$, find $f(u)$.

Sol. Let $u = e^x y$ ($u \neq 0$ since $y \neq 0$)

$$\frac{\partial z}{\partial x} = \frac{\partial f(u)}{\partial u} \cdot \frac{\partial u}{\partial x} = e^x y \frac{\partial f(u)}{\partial u}$$

$$\frac{\partial z}{\partial y} = \frac{\partial f(u)}{\partial u} \cdot \frac{\partial u}{\partial y} = e^x \frac{\partial f(u)}{\partial u}$$

$$e^x y \frac{\partial f(u)}{\partial u} + e^x \frac{\partial f(u)}{\partial u} = 1$$

$$zu \frac{df(u)}{du} = 1$$

$$\frac{df(u)}{du} = \frac{1}{zu}$$

$$f(u) = \frac{1}{2} \ln|u| + C$$

$$u = 1 \quad f(1) = C = 0 \Rightarrow C = 0$$

$$f(u) = \frac{1}{2} \ln|u|.$$

10. (8 pts) Find the Taylor series for $f(x) = \ln(x + \sqrt{x^2 + 1})$ at $x = 0$.

$$\text{S.I.} \quad f'(x) = \frac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} = \frac{1}{\sqrt{1+x^2}} \frac{\sqrt{1+x^2} + x}{x + \sqrt{1+x^2}} = \frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-\frac{1}{2}}$$

$$f'(x) = 1 + \sum_{k=1}^{\infty} \left[\begin{matrix} -\frac{1}{2} \\ k \end{matrix} \right] x^{2k} \quad (|x| < 1)$$

By Term-by-Term Integration Theorem,

$$f(x) = x + \sum_{k=1}^{\infty} \frac{1}{2k+1} \left[\begin{matrix} -\frac{1}{2} \\ k \end{matrix} \right] x^{2k+1} + C \quad (-1 < x < 1)$$

$$f(0) = \ln(0 + \sqrt{0^2 + 1}) = 0$$

$$f(0) = 0 + C = 0$$

$$\text{So } f(x) = x + \sum_{k=1}^{\infty} \frac{1}{2k+1} \left[\begin{matrix} -\frac{1}{2} \\ k \end{matrix} \right] x^{2k+1} \quad -1 < x < 1.$$